# Development of Predictive Models for Ophthalmologic Data Structures With Binary Outcomes

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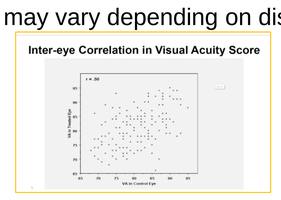
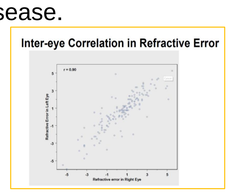
## A. Background

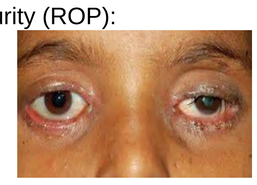
- The application of appropriate statistical strategies allows researchers to realize the full potential in analyzing data from paired ocular measurements, including those from prospective studies, and Genome-wide Association Studies (GWAS) [1].
- The nature of ocular measurements of paired eyes, however, pose challenges to ophthalmic researchers in developing predictive models for joint inference on paired-eye data[1].

- **Correlation in Ocular Data.**
  - **Positively correlated**: finding in one eye is likely to be more similar to that in the fellow eye of the same subject than to that in eye from different subject that to result from,
    - Environmental factors
    - Genetic factors
  - Inter-eye correlation may vary depending on disease.
    - Visual Acuity
    - Refractive Error

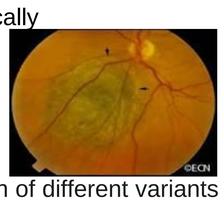



  - High correlation in retinopathy of prematurity (ROP):
    - > 80% are bilateral
  - Blepharitis, a chronic inflammation of the eyelid, almost always affect both eyes bilateral 95% of the time[2].



  - **Rare Disorders:** Rare conditions characteristically will occur in only one eye.
    - Choroidal melanoma, the most common primary malignant intraocular tumor, occurs in only one eye in > 98% of cases[3,4].



- The advent of GWAS has permitted the evaluation of different variants that contribute to the etiology of eye disorders- posing the dilemma of testing multiple hypotheses[1].
- In a perspective on application of advanced statistical methods in ophthalmology[1], authors noted that, when correlated measurements from both eyes (such as intraocular pressure or refractive error) are available, using the information from only one eye is a statistically simple approach, but may not reflect the true extent of the disease.
- The development and application of statistical methodology in modern ophthalmologic research is therefore vital to address these challenges.

## B. Motivation

- Though valuable knowledge are derived from analyses of the Age-Related Eye Disease Study (AREDS) trial, *methodological challenges remain* in evaluating risk factors for Age-related Macular Degeneration (AMD), the leading cause of blindness for persons >60 years and age-related cataract, the 3rd leading course of blindness in the US.
- In reviewing AREDS reported studies, we noted that much studies adopt simple analytic methods and regression models but did not account for the correlation between eyes.
- It is documented across a range of study designs, that when correlation between paired-eyes are ignored, the p-values and CIs are under-estimated, with consequence of inaccurate inference[5].
- There is also the situation where data is available on only one eye for some persons and on both eyes for others.
- The unique classification of case definition varies from diseases to disease; some requiring the presence of a clinical phenotype in one eye for case definition, others the presence in both eyes.
  - Classification of Category 4 AMD, require presence of Geographic Atrophy and/or Neovascular AMD in at least one eye
  - Severe glaucoma requires the presence of optic nerve damage with loss of vision in both eyes.
- These different facets of data structures and case definition pose challenges that require careful methodological approaches for efficient analysis and inference.
- Predictive models with the eye as the unit of analysis and deliberation of the correlation between paired-eyes offers optimal use of available data, and enhanced interpretability of risk factors.
- Of the 2415 DNA specimens available from the AREDS trial[6], 940 were from disease-free subjects and 1475 were from subjects with early or intermediate AMD in one or both eyes.
  - The overall distribution of disease occurrence in such data or similar outcomes should appropriately be a mixture.

## C. Objectives

- The **goal** of this project is to develop computationally tractable likelihood models for making risk prediction and inference for paired, unpaired and combined paired-unpaired data structures that commonly arise in ocular research, correcting for the correlation between eyes and allowing for inclusion of risk factors.

- The **objectives** are
  - **1**: To develop predictive models for making inference about ophthalmologic data with dichotomous outcomes that allows for inclusion of single-eye data, paired-eyes data and allow for covariate effects.
  - **2:** To develop models for the response probabilities in terms of mixed effects linear models within the framework of clustering accounting for correlation structures.
  - **3:** To develop Machine Learning (ML) algorithm using Maximum Likelihood Estimation (MLE) methods for parameter estimation, prediction and inference.

## D. Modeling Consideration[7]

- Suppose data are composed of N clusters, each of size $n_i$, i=1,…,N and a vector of binary responses $Y_i = (Y_{i1},\ldots,Y_{ini})^T$ measured on it.
- Let $Y = (Y_1, Y_2,, Y_N)^T$, then the $Y_i$'s are independently distributed vectors.
- Let $\pi_0$ and $\pi_1$ represent two latent classes where:
  - $\pi_0$ is a class (of families) whose members do not manifest the attribute under study
  - $\pi_0$ is called the zero-vector state.
  - $\pi_1$ is a class (of families) whose members are susceptible to the attribute and assume to have a Bernoulli distribution with prob. $\delta ij$.
  - $\pi_1$ is called the Bernoulli state.
- Define the unobserved random variable, $Z_i$, ($Z_i$=0,1); i=1,…N, such that

$$Z_i = \begin{cases} 0, & \text{with probability } 1-\alpha_i \\ 1, & \text{with probability } \alpha_i \end{cases}$$

- Suppose that $Z_i = 0$ when $\mathbf{Y}_i$ is generated from the **zero-vector state** and $Z_i = 1$ when $\mathbf{Y}_i$ comes from the **Bernoulli state**.
- Then

$$Y_{ij} = \begin{cases} 0, & \text{with probability } 1-\alpha_i \\ \mathrm{Ber}(\delta_{ij}), & \text{with probability } \alpha_i \end{cases}$$

- Then for the $j^{th}$ outcome in the $i^{th}$ cluster

$$\delta_{ij} = P(Y_{ij} = 1 \mid Z_i = 1)$$

- It is shown[7] that the joint probability distribution of the $i^{th}$ cluster is:

$$P(Y_{i1} = y_{i1},\ldots,Y_{in_i} = y_{in_i}) = (1-\alpha_i)\prod_{j=1}^{n_i}(1-y_{ij}) + \alpha_i \prod_{j=1}^{n_i}\delta_{ij}^{y_{ij}}(1-\delta_{ij})^{1-y_{ij}}$$

- This is a mixture of a form of a degenerate distribution representing the class that does not manifest the attribute and an independent distribution representing the class that yield correlated outcomes.

### D.1 Features of the Joint Distribution

- Several desirable properties of the joint distribution are discussed[7].
- Of note is that the joint distribution is reproducible.
  - That is, the marginal distribution of any subset of $Y_1,\ldots,Y_n$ has the same form as the joint distribution.
    - The interpretation of the parameters is the same regardless of cluster size.

### D.2 Parametrization for Risk Prediction

- Suppose the $j^{th}$ subject has a vector of $\boldsymbol{p}$ individual covariates, $\mathbf{X}_{ij} = (x_{ij1},\ldots,x_{ijp})$
- Let $Z_i$ ($Z_i$=0,1) ,i=1,…,N be an unobserved random variable.
- The scientific objective is to characterize the dependence of $\mathbf{Y}_{ij}$ on $\mathbf{X}_{ij}$.
- The observations may be taken to be independent between clusters, but certain within-cluster and within-individual correlations are anticipated, which can be modeled explicitly through random effects attached to the linear predictors.
- Therefore, to account for excess unobserved variation, not accounted for by cluster-specific covariates, we introduce a random-effect term.

$$\delta_{ij}(\mathbf{X}) = P(Y_{ij} = 1 \mid Z_i = 1, \mathbf{X}_{ij}) = \frac{1}{1+\exp[\beta_0 + \beta X_{ij} + \gamma_i]} \quad \text{where } \gamma_i \sim N(0,\sigma^2)$$

### D.2 Maximum Likelihood Estimation for Machine Learning Algorithm

- Let $\boldsymbol{\theta} = \{\alpha, \beta_0, \boldsymbol{\beta}, \sigma^2\}$ be the parameters to be estimated
- MLE provides a framework for **predictive modeling** in **ML** where finding model parameters is framed as an optimization problem.
- The **condition distribution** of $\mathbf{Y} = (Y_1,.,Y_N)^T$, given $\gamma = (\gamma_1,,\gamma_N)$ is:

$$P(\mathbf{Y} \mid \gamma, \theta) = \prod_{i=1}^{N}\left\{(1-\alpha_i)\prod_{j=1}^{n_i}(1-y_{ij}) + \alpha_i \prod_{j=1}^{n_i}\delta_{ij}^{y_{ij}}[1-\delta_{ij}]^{1-y_{ij}}\right\}, \ \gamma_i \sim N(0,\sigma^2)$$

- Applying the M-point Gaussian quadrature (*numerical integration*), the **log-likelihood of equation** of the (marginal) distribution of **Y** – found by integrating out the conditional distribution *wrt* γ - is derived as

$$\log L(\boldsymbol{\beta}, U \mid \mathbf{Y})) \approx \sum_{i=1}^{N}\log\left\{(1-\alpha_i)\prod_{j=1}^{n_i}(1-y_{ij}) + \frac{\alpha_i}{\sqrt{\pi}}\sum_{m=1}^{M}w_m\left[\prod_{j=1}^{n_i}(\delta_{ijm})^{y_{ij}}(1-\delta_{jim})^{1-y_{ij}}\right]\right\}$$

where,

$$\delta_{ijm}(\beta_0, \boldsymbol{\beta}, \sigma) = \frac{1}{1+\exp\{\beta_0 + \boldsymbol{\beta}X_{ij} + \sqrt{2}\sigma\upsilon_m\}}$$

and $(\omega_m, \upsilon_m) \sim$ *quadrature* (weights and points).

- MLE is a fundamental concept in machine learning and statistics that plays a crucial role in parameter estimation for probabilistic models. It forms the backbone of many algorithms and techniques used in modern data-driven applications[8].
- Section E discusses application of the modeling consideration to ocular data

## E. Application to Dichotomous Ocular Data

- Consider a collection of N individuals each with a pair of "eligible" eyes under study.
- Let $i = 1,2,\ldots N$ index the $i^{th}$ individual
- Let $j = [L,R]$ , ~ (L=Left, R=Right) eye.
  - The individual represents a clustering unit
  - The eye the unit of analysis.

- For the $i^{th}$ individual, define $Y_{ij}$, such that

$$Y_{iL} = \begin{cases} 1, & \text{if the left eye is diseased} \\ 0, & \text{if the left eye is not diseased} \end{cases} \quad \text{and} \quad Y_{iR} = \begin{cases} 1, & \text{if the right eye is diseased} \\ 0, & \text{if the right eye is not diseased} \end{cases}$$

- Then the disease vector, $D_i$, for the $i^{th}$ individual is

$$D_i = [y_{iL}, y_{iR}] = [(0,0),(0,1),(1,0),(1,1)], \quad i = 1,\ldots,N$$

- For the $i^{th}$ person, let

$$\alpha_i = \frac{P(Y_{iL}=1)P(Y_{iR}=1)}{P(Y_{iL}=1, Y_{iR}=1)}, \quad i = 1,2,\ldots,N$$

- If $\boldsymbol{\alpha_i = 1}$, then the outcomes on the eyes are (*independent*) randomly determined
  - that is there is no correlation (*dependence*) between eyes.
  - thus, $\boldsymbol{\alpha_i}$ is a measure of the between eye correlation.
- Let,

$$\delta_{ij} = P(Y_{ij}=1 \mid Y_{ij'}=1); \ j \neq j' = [L,R]$$

  - Then $\delta_{ij}$ is simply the conditional probability of the outcome on one eye, given the fellow eye has the attribute.
- With the above we show that the joint probability distribution for the $i^{th}$ person can be based on

$$P(Y_{iL}, Y_{iR}) = (1-\alpha_i)\prod_{j=[L,R]}(1-y_{ij}) + \alpha_i \prod_{j=[L,R]}\delta_{ij}^{y_{ij}}(1-\delta_{ij})^{1-y_{ij}}$$

  - The joint distribution is reproducible.
    - Marginal distribution of an eye has same form as the joint distribution of the pair of eyes.
    - Interpretation of the parameters is the same regardless of no. of eligible eyes.
    - Hence the situation where a person presents data on a single-eye or paired eyes causes no problem should this approach be adopted.
- Let

$$\boldsymbol{\mu_{ij} = P(Y_{ij} = 1), j = [L,R]}$$

  - Then $\boldsymbol{\mu_{ij}}$ is simply a measure of the population disease rate.
  - It follows that $\delta_{ij} = \mu_{ij}/\alpha_i$ .
- And so, an alternative parameterization of the joint distribution with interpretable parameters in terms of $\boldsymbol{\mu_{ij}}$ and $\boldsymbol{\alpha_i}$ is given as

$$P(Y_{iL}, Y_{iR}) = (1-\alpha_i)\prod_{j=[R,L]}(1-y_{ij}) + \alpha_i \prod_{j=[L,R]}\left(\frac{\mu_{ij}}{\alpha_i}\right)^{y_{ij}}\left(1-\frac{\mu_{ij}}{\alpha_i}\right)^{1-y_{ij}}$$

- With this parameterization, we can model the mean, $\boldsymbol{\mu_{ij}}$, directly on covariates, **X**, without having to stratify the analysis of response of one eye on the other eye.

## Planned & Ongoing Studies

- Data analysis and reporting step is our ultimate goal of this project.
- We will initially utilize the comprehensive dataset from AREDS, a well-designed prospective clinical trial with rich phenotype, and clinical information as well as genetic data.
  - We have applied and granted access to the AREDS Database through dbGaP, an NIH database of Genotypes and Phenotypes.
  - In the process of extracting relevant data for application
- **We plan to apply the model analyze data to**
  - Examine association of environmental and genetic effects and their interaction with the development of ADM correcting for the correlation between paired eyes
  - Examine the association of environmental and genetic factors and their interaction with different cataract phenotypes (nuclear only, cortical, posterior sub-capsular and mixed) correcting for correlation between paired eyes.

## Future Directions

- Measurement of multivariate outcomes occur time and again in ophthalmologic research, usually because the diseases are related or form a constellation as a syndrome.
- It is often of clinical interest to model the inter-correlation not only between an outcome and risk factors, but also between different outcomes.
- In the AREDS trial, both AMD and age-related cataract, the two primary and related disorders were assessed. However, the bivariate nature of the outcomes has not been considered in any AREDS study and applied methods especially for dichotomous outcomes are limited. For future studies, we will explore different predictive models for the simultaneous assessment of association between the bivariate outcomes of ADM and cataract with both environmental and genetic factors.
- A common feature in the study of clusters is the existence of multiple groups with levels of nesting within it.
  - For example, Benson et al[9] studied patients with retinitis pigmentosa, an eye disease in families. Here subjects are nested within families and two eyes are nested within each subject.
  - Thus, there are family-specific, patient-specific, and eye-specific covariates.
  - For future studies, we will develop an extension of our proposed models to two-way hierarchical layout.

## Reference

1. Fan, Q., Teo, Y. Y., & Saw, S. M. (2011). Application of advanced statistics in ophthalmology. *Investigative ophthalmology & visual science*, *52*(9), 6059-6065.
2. Castella et al. (1995). Study of the blood-aqueous barrier in choroidal melanoma. *British journal of ophthalmology*, *79*(4), 354-357.
3. Dunn et al. (1988). Choroidal malignant melanoma with early vitreous seeding. *Retina*, *8*(3), 188-192.
4. Seal et al. (1995). Placebo controlled trial of fusidic acid gel and oxytetracycline for recurrent blepharitis and rosacea. *British journal of ophthalmology*, *79*(1), 42-45.
5. Glynn, R. J., & Rosner, B. (2012). Regression methods when the eye is the unit of analysis. *Ophthalmic epidemiology*, *19*(3), 159-165.
6. Perlee et al. (2013). Inclusion of genotype with fundus phenotype improves accuracy of predicting choroidal neovascularization and geographic atrophy. *Ophthalmology*, *120*(9), 1880-1892.
7. Kwagyan et. Al (2018). Modeling clustered binary data with excess zero clusters. *Statistical methods in medical research*, 27(9), 2641-2656.
8. Brown, J (2019), A Gentle Introduction to Maximum Likelihood Estimation for Machine Learn found in https://machinelearningmastery.com/what-is-maximum-likelihood-estimation-in-machine-learning/

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